Ahmed Bennasser

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LinkedIn • GitHub • Portfolio

Computer Science and Applied Mathematics engineering student at ENSIMAG, focused on Machine Learning, Statistical Learning, and NLP, with a particular interest in LLMs and Generative AI. Experience in fine-tuning, Transformers, RAG, and model evaluation.
I am seeking a 5 to 6-month End-of-Study internship in AI/ML starting February 2027.

EDUCATION

2024 – Present — **Grenoble INP – Ensimag & Université Grenoble Alpes**

- **Master 2 MSIAM** – MSc in Industrial and Applied Mathematics (2026–2027)
- **Engineering Degree** – Mathematical Modeling, Image and Simulation track.

2022 – 2024 — **Lycée Mohammed VI d'Excellence**
Intensive two-year undergraduate program in Mathematics & Physics (CPGE)

EXPERIENCE

June 2026 – July 2026 **Assistant Engineer Internship – LIG / TIMC, Grenoble**

- Reimplementation of the *Hierarchical Feature Engineering* method leveraging taxonomic structure on sparse count data (microbiome data).
- Design and implementation of a statistical algorithm (**SFE**) for feature selection, evaluated against **HFE** and reference models across 5 experimental protocols: consistency, within-study, cross-study, noise stability, and high-dimensional regime ($p \gg n$).
- **Technologies:** Python, Scikit-learn, NumPy, Pandas, Statistical Analysis, Feature Engineering.

February 2026 – May 2026 **Research Project – LIG (GETALP Team), Grenoble**

- Analysis and evaluation of the French multimodal encoder *Pantagruel* (LIG) on NLP and NER tasks.
- Design of a text-to-pictogram generation pipeline: paired dataset creation, fine-tuning of diffusion models via **LoRA**, **Prompt Engineering** and **inference**.
- Development of a hybrid evaluation framework combining mathematical style and contrast metrics with semantic validation (***LLM**-as-a-Judge*).
- **Technologies :** Python, NLP, NER, Transformers, PEFT, Quantization, JEPA, Word2Vec, LoRA, Stable Diffusion, Reward Models, LLM-as-a-Judge, Prompt Engineering. GitHub

PERSONAL PROJECTS

June 2026 – Present **AgriGuard-RAG — Agentic RAG for Phytosanitary Regulations** **Live Demo**

- Developed an Agentic RAG system querying 5 277 official French phytosanitary (E-Phy) regulations, deployed in production on **Streamlit Cloud**.
- Designed a hybrid router combining rule-based heuristics and **TF-IDF** + **Logistic Regression** to dynamically classify query complexity and intent.
- Built a **hybrid retrieval engine** on **Qdrant Cloud** combining dense embeddings (FastEmbed), sparse **BM25**, and metadata filtering with **Reciprocal Rank Fusion (RRF)**.
- Implemented an **Agentic reasoning workflow** via **Groq Cloud API**: autonomous query decomposition into sub-queries, parallel vector retrieval, context deduplication, and zero-hallucination synthesis.
- **Technologies:** Python, **Qdrant Cloud**, **Groq API**, FastEmbed, BM25, Scikit-learn, Streamlit, **LLM**. **GitHub**

November 2025 **Climate Sentiment Analysis**

- Sentiment analysis applied to texts from Twitter related to climate issues.
- Comparative benchmark (RNN, LSTM, BERT): performance analysis (F1-score, precision, recall) illustrating BERT's advantage on large datasets and LSTM's robustness on limited data.
- Technologies: PyTorch, BERT, RNN, LSTM, Hugging Face Transformers, Pandas. GitHub

September 2025 **Chest X-Ray Medical Diagnosis**

- Classification of more than 20 000 medical images into four categories, with data augmentation and the use of CNNs as well as Transfer Learning methods based on ResNet18 and EfficientNet.
- Comparative evaluation of ResNet-18, EfficientNet, and a CNN developed from scratch, highlighting the performance gains provided by Transfer Learning. GitHub

SKILLS & LANGUAGES

Languages & Tools
Python | Java | C | C++ | SQL | R | Git | Linux | Jupyter | VS Code | LaTeX

AI & Machine Learning & Data Science
PyTorch | Scikit-learn | LLM | Transformers | Hugging Face | Qdrant | LangChain/LlamaIndex | BM25 | NumPy | Pandas | Feature Engineering | Data Analysis | Prompt Engineering | Multimodal models

Arabic: Native | **French:** Fluent | **English:** Professional (scientific reading, technical writing)